

GCN-BASED FUNCTIONAL PREDICTION OF UNCHARACTERIZED PROTEINS

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INTRODUCTION

Protein function prediction is one of the most important challenges of the postgenomic era, because of the exponential increase in sequenced proteins and the **slower experimental characterization**.¹

Computer models have been used for Automatic Function Prediction (AFP) for more than 20 years, but the recent adoption of Deep Learning has opened a new paradigm, with modern frameworks like MVGCN integrating multiple "views" of data, helping to detect unobserved links in graph networks.² These new models are used for Drug-Target Interaction (DTI) prediction, miRNA-disease association identification, and *de novo* molecular design.^{2,3}

AFP models are particularly suited for the **study of organisms where not all proteins are characterized**, mostly due to the barebones functional analysis carried out when the organism is first isolated and studied.¹ In most cases the experimental studies are focused just on 1 or 2 proteins and the rest are given a cursory prediction of the CDS's function and not followed up. One of these organisms is *Haloquadratum walsbyi*, with 419 uncharacterized proteins out of 2558.⁴

HYPOTHESIS AND OBJECTIVES

Hypothesis:

Given the state of GCN technology, **functional prediction of a series of proteins is accurate enough to assign a putative function.**

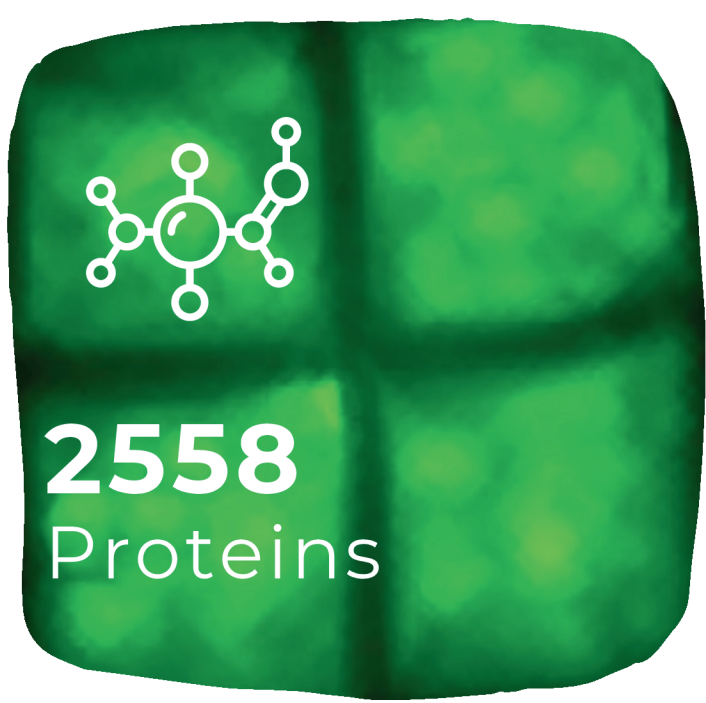
Objectives:

To obtain **new information from an understudied genome.**

- To **identify protein function** using GCN-based bioinformatics tools.
- To **give context to this information** by comparing it to various databases.

METHODOLOGY

Haloquadratum walsbyi DSM16790



419
Uncharacterized
Proteins

26
Studied
Proteins

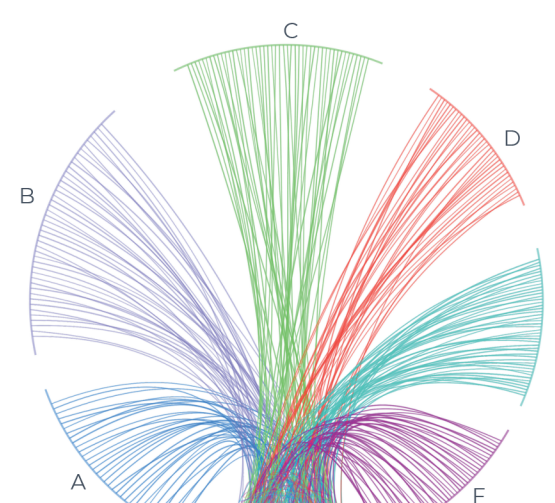
I-TASSER
Protein Structure & Function Predictions

**Putative Function
Assignment**

National
Center for
Biotechnology
NCBI
Information

BLASTn and BLASTp

1 run exclusive for *Haloquadratum*
1 run excluding *Haloquadratum*



**Clusters of
Orthologous Genes**



KEGG Orthology

Figure 1. Schematic depiction of the used methodology.

RESULTS AND DISCUSSION

Figure 2. Graphical representation of the results obtained.

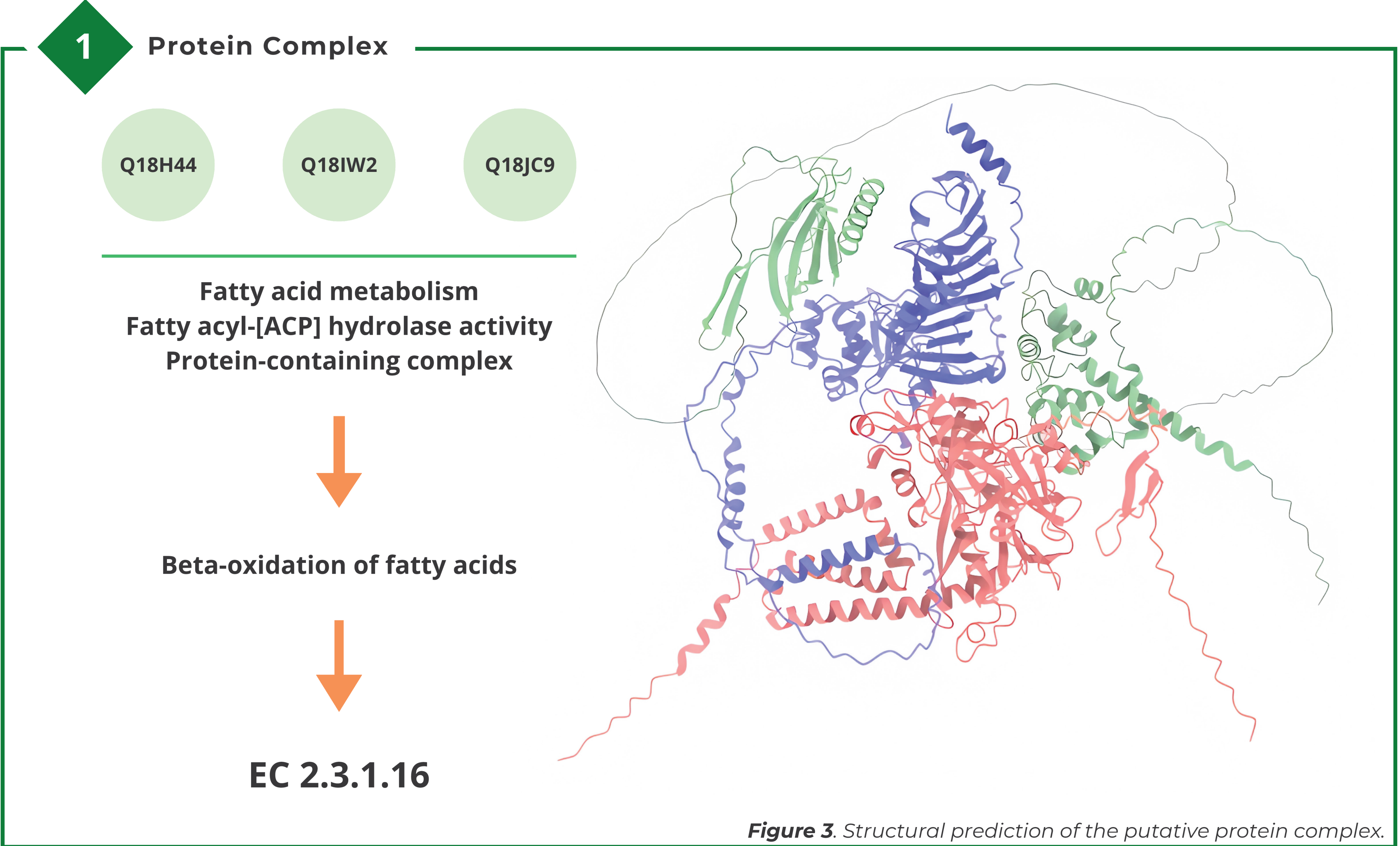
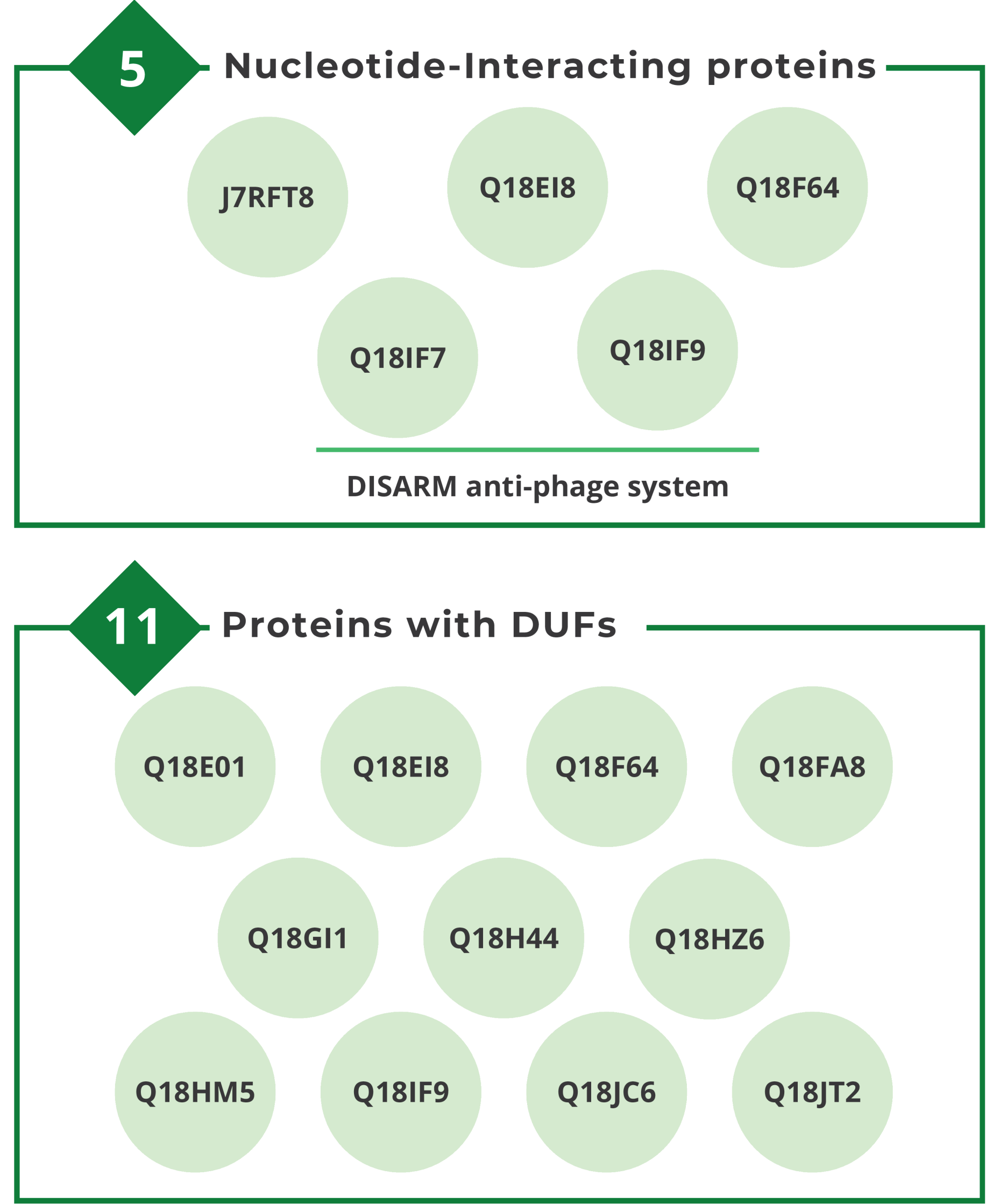


Figure 3. Structural prediction of the putative protein complex.

CONCLUSIONS

This study **assigned a putative function to a series of uncharacterized proteins**, whose functions will remain putative until experimental data for the individual proteins or their domains can be gathered. **Expanding the knowledge about Domains of Unknown Function and their functions will be the key to increased prediction accuracy** in the future because some proteins will just be assigned a function via homology. Finally, this study has found **GCNs to be a valid tool for protein function prediction**, and as the information pool grows **their predictions will eventually be used in the same likeness as metagenome-assembled genomes.**

REFERENCES

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SCAN ME

